

Drive train dynamics assessment and speed controller design in variable speed wind turbines



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ABSTRACT

This paper deals with the speed controller design in DFIG based wind turbines, and investigates stability and performance of the drive train dynamics against different control strategies. It is shown that speed controller design based on the single mass drive train model may result in unstable mechanical modes, because it ignores the dynamics of the flexible shaft. Then, another control approach, known as feed-forward compensation of the shaft torsional torque, is examined. It is shown that this control method results in poorly damped oscillations of torsional torque and turbine speed during the transient conditions. The open loop transfer function from the electromagnetic torque to the generator speed contains a dual quadratic function representing the dynamics of flexible shaft. The dual quadratic function comprises resonant and anti-resonant frequencies that greatly affect the stability of the drive train dynamics. Next, a step-by-step procedure for designing the speed controller based on the two-mass drive train model is proposed. The proposed speed controller provides stable closed loop system, zero tracking error, low-frequency disturbance rejection, and open-loop gain attenuation at the resonant frequency. At the end, performance of the proposed controller is investigated by the time domain simulations.

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1. Introduction

Wind turbine based on the DFIG is a well-known configuration extensively used in the wind turbine industry. The DFIG-based wind turbine, shown in Fig. 1, consists of drive train system, induction generator, back-to-back voltage source converters and control system. The back-to-back voltage source converters comprise rotor-side and grid-side converters. The rotor-side converter (RSC) is used to control the generator speed and reactive power, while the grid-side converter (GSC) is used to control the dc-link voltage and reactive power exchanged with the grid [1].

The wind turbine equivalent shaft is relatively softer than the typical turbine shaft in conventional power plants [2]. Hence, the drive train of the wind turbine should be described with at least a two-mass model [3], or three-mass model [4], since a single-mass shaft model cannot properly represent mechanical oscillations.

In Ref. [5] the transient stabilities of a three-mass model, a two-mass model, and a one-mass model are compared. Refs. [6,7] conclude that a two-mass drive train model is sufficient for transient stability analysis of wind turbine-generator systems. In a two-

mass model the low speed mass of the turbine is connected to the high speed mass of the generator through a flexible shaft modeled as a spring and damper.

This paper deals with the performance study and stability analysis of the drive train dynamics together with consideration of the elaborate speed controller design in DFIG based wind turbines.

In the DFIG, the RSC controls the generator to achieve maximum power point tracking. In one classification, the control approaches of the DFIG are categorized in two groups: classic control approaches using PI regulators and advanced control approaches. In Refs. [8–13] advanced regulators based on the nonlinear control approaches are introduced for the DFIG control. In another classification, in literatures, the DFIG controls are mainly divided into two categories: direct control of the generator known as direct torque control (DTC) and direct power control (DPC), and indirect control known as field oriented control (FOC). The direct control remedies of DTC and DPC, given in Refs. [14,15], have the ability of directly controlling the generator torque and power through properly selecting the inverter switching states [16]. The field oriented control, as a well-known control strategy, provides the ability of separately controlling the active and reactive power of the generator and is widely used in the DFIG control designs [17–19]. The FOC scheme is implemented based on two control loops: inner

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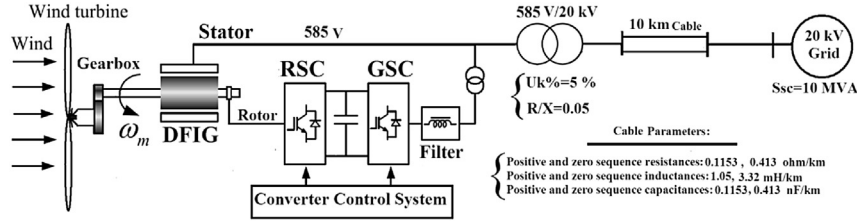


Fig. 1. DFIG system connected to the grid.

current loop and outer control loop using PI controller. In FOC, the rotor current, as an interface variable, is used to control the active and reactive power of the DFIG. From the outer control loop point of view, there are two modes for the FOC implementation: power control mode [20] and speed control mode [21]. The reference control value is stator power for the power control mode [22,23], and the rotational speed for the speed control mode [19,21]. Both the speed and power control modes are working to maximize the power output of the wind turbine across a wide range of wind speeds by adjusting the turbine rotational speed or generator power according to the optimal tracking characteristic, which considers the rotor blade characteristics. However, the power and speed control have differences in dynamic characteristics [24]. The power control mode has slow response with simple construction, while the speed control mode has fast response with complex construction [25].

In most of the research works regarding the DFIG control, the drive train has been modeled as a single mass model. However, little work has been published regarding analytical studies and controller design of the DFIG by considering multi mass drive train model. For the DFIG control, it is required to take into account the multi mass drive train dynamics. This is because torsional oscillations associated with the drive train natural frequency significantly affect the wind turbine performance resulting fatigue and stress on the mechanical system.

As the main contribution, this paper develops theoretical analyses for the elaborate speed controller design in DFIG based wind turbines using two-mass drive train model to take into account the torsional dynamics associated with the multi mass drive train model. Further, the paper examines the stability and dynamic performance of the wind turbine against different control strategies by considering the multi mass drive train model. In this way, first, the speed controller design based on the single mass drive train model is given, and dynamic performance of the system is examined using the modal and frequency response analyses. It is shown that this control method may result in weakly damped mechanical modes for the lower control bandwidths and even unstable mechanical modes for the higher control bandwidths.

Then, another control approach based on the feedforward compensation of the shaft torsional torque is evaluated. It is shown that this control method results in poorly damped oscillations of the torsional and electromagnetic torques during the transient conditions.

Next, in order to damp the low-frequency oscillations of the drive train, the speed controller design based on the two-mass drive train model is proposed. The proposed speed controller provides stable closed loop system with adequate stability margin, zero tracking error, better low-frequency disturbance rejection, and open-loop gain attenuation at the resonant frequency. At the end, performance of the proposed controller is examined by the time domain simulations.

2. DFIG modeling and control

In the following the dynamic equations of the DFIG are proposed, in which all system parameters and variables are in per unit, and positive direction for the stator and rotor currents is assumed into the generator.

The stator and rotor voltages, fluxes, and electromagnetic torque in a synchronous reference frame rotating at angular speed of ω_s are given by Ref. [26]:

$$v_{sdq} = R_s i_{sdq} + j\omega_s \psi_{sdq} + \frac{1}{\omega_b} \frac{d\psi_{sdq}}{dt} \quad (1)$$

$$v_{rdq} = R_r i_{rdq} + j\omega_2 \psi_{rdq} + \frac{1}{\omega_b} \frac{d\psi_{rdq}}{dt} \quad (2)$$

$$\begin{aligned} \psi_{sdq} &= L_s i_{sdq} + L_m i_{rdq} \\ \psi_{rdq} &= L_m i_{sdq} + L_r i_{rdq} \end{aligned} \quad (3)$$

$$T_e = \frac{L_m}{L_s} (\psi_{sq} i_{rd} - \psi_{sd} i_{rq}) \quad (4)$$

where ψ , v and i represent the flux, voltage and current. Subscripts s and r denote the stator and rotor quantities, respectively. L_s and L_r are the stator and rotor self inductances, L_m is the mutual inductance. ω_2 is the rotor slip frequency, ω_b is the base angular frequency, and ω_s is the speed of d–q synchronous reference frame. Also, R_s and R_r are the stator and rotor resistances.

In the dq reference frame with the stator voltage orientation (SVO), the d-axis is aligned with the stator voltage vector, and consequently $v_{sd} = V_s$ and $v_{sq} = 0$. With stator and rotor resistances neglected, the steady state stator/rotor voltage equations in SVO frame are obtained as follows: $V_s = v_{sd} \approx -\omega_s \psi_{sq}$, $V_{sq} \approx \omega_s \psi_{sd} = 0$ and $V_r \approx \frac{L_m}{L_s} s V_s$, where ω_s , equal to 1 pu, is the stator angular frequency, s is the generator slip, and V_s is the stator voltage amplitude. This results in the approximate expressions of $\psi_{sd} \approx 0$ and $\psi_{sq} \approx -|\psi_s|$.

Considering (4) and according to the explanations given above, the electromagnetic torque can be written as

$$T_e = \frac{L_m}{L_s} (\psi_{sq} i_{rd}) \approx -\frac{L_m}{L_s} (|\psi_s| i_{rd}) \quad (5)$$

According to (5), in SVO frame, the electromagnetic torque and consequently the generator speed can be controlled by the d-axis rotor current i_{rd} . On the other hand, generator voltage and reactive power can be controlled by the q-axis rotor current i_{rq} .

2.1. Rotor current control loop

From (2) and (3), the rotor dynamics is described in terms of rotor current and stator flux, as follows [27].

$$v_{rdq} = R'_r i_{rdq} + \frac{L'_r}{\omega_b} \frac{di_{rdq}}{dt} + j\omega_2 L'_r i_{rdq} + e_{dq} \quad (6)$$

where the terms R'_r and L'_r in (6) are transient resistance and inductance of the rotor current dynamics, and are given by:

$$R'_r = R_r + \left(\frac{L_m}{L_s}\right)^2 R_s \quad (7)$$

$$L'_r = L_r - \frac{L_m^2}{L_s}$$

Also, the term e_{dq} in (6) is the d–q rotor back-emf voltage induced in the rotor winding and reflects the effects of stator dynamics on rotor current dynamics. It is a function of stator flux and voltage, and is given by:

$$e_{dq} = \frac{L_m}{L_s} \left(v_{sdq} - j\omega_r \psi_{sdq} - \frac{R_s}{L_s} \psi_{sdq} \right) \quad (8)$$

where ω_r in (8) is the rotor speed.

Considering the rotor current controller to be PI, $K_{I_r}(s) = K_{P-I_r} + K_{I-I_r}/s$, the following rotor current control loops, given in Fig. 2, are obtained. The rotor d–q voltages, v_{rd} and v_{rq} , act as control inputs determined by the rotor current controller.

According to Fig. 2, the open loop bandwidth of the rotor current control is $\alpha_s = \frac{R'_r}{L'_r} \omega_b$, which is relatively small. By canceling the plant pole with the zero of the controller, in Fig. 2, we have

$$\frac{K_{I-I_r}}{K_{P-I_r}} = \frac{R'_r}{L'_r} \omega_b \quad (9)$$

Selecting $K_{P-I_r} = \frac{\alpha_l L'_r}{\omega_b}$, and taking (9) into account, the transfer function from $i_{rdq-ref}$ to i_{rdq} will be

$$G_{cl-I_r} = \frac{I_r(s)}{I_{r-ref}(s)} = \frac{\alpha_l}{s + \alpha_l} \quad (10)$$

where α_l is the closed loop bandwidth of the rotor current control, $\alpha_l = \frac{K_{P-I_r}}{L'_r} \omega_b$.

2.2. Speed controller design based on the single-mass drive train model

For operating the wind turbine at maximum power point tracking (MPPT) mode, it is required that the generator tracks the optimal reference speed through the speed control loop. In this section for the simplicity the wind turbine speed controller is designed based on the single-mass drive train model. The single-mass drive train model of the wind turbine is given by

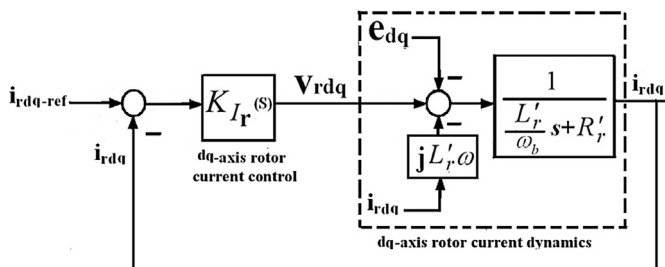


Fig. 2. Rotor current control loop.

$$2H \frac{d\omega_r}{dt} = T_e + T_t \quad (11)$$

where H is the total inertia constant of turbine-generator in (sec), ω_r is the generator speed in (pu), T_e and T_t are generator electromagnetic torque and turbine mechanical torque in (pu), respectively. Fig. 3 shows the speed control scheme of the wind turbine. ω_{r-ref} in Fig. 3 is the generator reference speed, and is obtained to have optimal tip-speed ratio, also α_l is the bandwidth of the d-axis rotor current control loop, and $K_\omega(s)$ is the PI speed controller, $K_\omega(s) = k_{P-\omega} + k_{I-\omega}/s$. The gain $\frac{L_m}{L_s} |\psi_s|$ in Fig. 3 is approximately equal to 1 at steady state, and can be replaced with 1. If the controller parameters $k_{P-\omega}$ and $k_{I-\omega}$ are selected such that

$$\frac{k_{P-\omega}}{2(H_g + H_t)} < \alpha_l \quad (12)$$

$$\frac{k_{I-\omega}}{k_{P-\omega}} < \alpha_l$$

then under the frequency range of $\omega \ll \alpha_l$, we have

$$\frac{\omega_r}{\omega_{r-ref}} = \frac{2\xi\omega_n s + \omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2} \quad (13)$$

where ω_n and ξ are the undamped natural frequency and damping ratio of the speed control loop, given by

$$\omega_n^2 = \frac{k_{I-\omega}}{2(H_g + H_t)} \quad (14)$$

$$2\xi\omega_n = \frac{k_{P-\omega}}{2(H_g + H_t)}$$

In other words, ω_n is the bandwidth of the speed control loop. If ω_n is selected small enough with respect to α_l , the inequalities of (12) are satisfied. The state equation corresponding to speed controller is given by

$$\frac{dx_4}{dt} = (\omega_{r-ref} - \omega_r) \quad (15)$$

Since the rotor current control loop is fast enough compared to the speed control, it is possible to set T_e to its reference value T_{e-ref} . Hence, T_e as a function of the generator speed ω_r is given by

$$T_e = k_{P-\omega} \cdot (\omega_{r-ref} - \omega_r) + k_{I-\omega} \cdot x_4 \quad (16)$$

3. Analysis of drive train dynamics with two-mass drive train model

The drive train of the wind turbine is usually represented by a two-mass model. In the two-mass model, the low speed mass of the turbine is connected to the high speed mass of the generator through a flexible shaft modeled as a spring and damper. The linearized model of the two-mass drive train is given by Ref. [28].

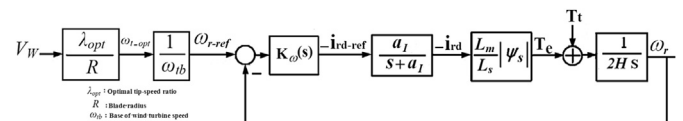


Fig. 3. Wind turbine speed control loop based on the single mass drive train model.

$$\frac{d\Delta\omega_r}{dt} = \frac{\Delta T_e + \Delta T_{sh}}{2H_g} \quad (17)$$

$$\frac{d\theta_{sh}}{dt} = \omega_b(\Delta\omega_t - \Delta\omega_r) \quad (18)$$

$$\frac{d\Delta\omega_t}{dt} = \frac{\Delta T_t - \Delta T_{sh}}{2H_t} \quad (19)$$

$$\Delta T_{sh} = k_s \Delta\theta_{sh} + D_{tg}(\Delta\omega_t - \Delta\omega_r) \quad (20)$$

where ω_t and ω_r are the turbine and generator speeds in (pu), θ_{sh} is the shaft twist angle in (rad), H_g and H_t are the inertia constants of the generator and turbine in (sec), respectively, k_s is the shaft stiffness coefficient in (pu/elec. rad), D_{tg} is the damping coefficient of the shaft in (pu), T_e and T_t are the generator electrical torque and the turbine mechanical torque, respectively, in (pu), and T_{sh} is the shaft torsional torque, in (pu).

3.1. Frequency response analysis

Considering (17)–(20), the open loop transfer function from the electromagnetic torque, ΔT_e , to the generator speed, ω_r , is given by

$$\frac{\Delta\omega_r}{\Delta T_e} = -\frac{1}{2(H_g + H_t)s} \cdot \frac{2H_t s^2 + D_{tg}s + k_s\omega_b}{\frac{2H_g H_t}{H_g + H_t} s^2 + D_{tg}s + k_s\omega_b} \quad (21)$$

where $\omega_{01} = \sqrt{\frac{k_s\omega_b(H_g + H_t)}{2H_g H_t}}$ is the resonant frequency, $\xi = \frac{D_{tg}}{2} \sqrt{\frac{1}{k_s\omega_b} \frac{(H_g + H_t)}{2H_g H_t}}$ is the damping ratio, and $\omega_{02} = \sqrt{\frac{k_s\omega_b}{2H_t}}$ is the anti-resonant frequency.

The transfer function of (21) is product of two terms: the first one corresponds to the single mass model, and the second one represents the shaft twist angle dynamics. Hence, the lower shaft stiffness results in the lower resonant and anti-resonant frequencies and consequently the higher effect on the drive train stability.

For the DFIG system of Appendix A with $H_g = 0.75$ sec, $H_t = 4.3$ sec, $k_s = 0.6$ pu/elec. rad, $D = 1.2$ pu, and $\omega_b = 2\pi \times 60$ rad/sec, the resonant and anti-resonant frequencies are $\omega_{01} = 2\pi \times 2.1$ rad/sec and $\omega_{02} = 2\pi \times 0.81$ rad/sec, respectively.

Fig. 4 shows the bode diagram of $\Delta\omega_r/\Delta T_e$ transfer function for the single-mass and two-mass drive train models. According to Fig. 4, in the case of the two-mass model, two abrupt changes are

seen at the bode plot due to the resonant and anti-resonant frequencies.

Fig. 5 shows the wind turbine speed control loop based on the two mass drive train model. According to Fig. 5, the overall open loop transfer function of the speed control system is given by

$$G_{op-\omega}(s) = \left(k_{p\omega} + \frac{k_{i\omega}}{s}\right) \left(\frac{\alpha_{dq}}{s + \alpha_{dq}}\right) \left(\frac{L_m}{L_s} \psi_{sd0}\right) \times \left(\frac{1}{2(H_g + H_t)s} \frac{2H_t s^2 + D_{tg}s + k_{s-eq}\omega_b}{\frac{2H_g H_t}{H_g + H_t} s^2 + D_{tg}s + k_{s-eq}\omega_b}\right) \quad (22)$$

Fig. 6 shows the bode plot of (22) for the DFIG system with parameters of Appendix A and speed control bandwidth of 5 Hz. The speed control design is based on the single mass drive train model given in Section 2.2.

According to Fig. 6, the magnitude and phase of the open loop system rapidly varies around the anti-resonant and resonant frequencies. At the anti-resonant frequency, $\omega = 5.12$ rad/sec, the magnitude and phase of the open loop system reach the 0 Bb and 165 deg, respectively, representing a low stability margin.

Also, at the resonant frequency, $\omega_{01} = 13.2$ rad/sec, the magnitude of the bode plot is 50 dB, that result in significant amplification of the torque ripple at this frequency.

3.2. Modal analysis

In the following the modal analysis of the drive train dynamics with the designed speed controller of Section 2.2 is presented. Considering (17)–(20), (15) and (16), the linearized drive train dynamic may be written as

$$\Delta \dot{X} = A \cdot \Delta X \quad (23)$$

where A is the 4×4 state matrix, and ΔX is the vector of state variables, i.e. $\Delta x = [\Delta\omega_r, \Delta\theta, \Delta\omega_t, \Delta x_4]^T$. The studies are carried out on a test DFIG system with parameters of Appendix A. The drive train of the turbine-generator has the following parameters: $H_g = 0.75$ sec, $H_t = 4.3$ sec, $k_s = 0.6$ pu/elec. rad, $D = 1.2$ pu, and $\omega_b = 2\pi \times 60$ rad/sec.

Tables 1 and 2 show the system modes, and dominant state variables for the speed control bandwidths of $\omega_n = 2\pi \times 2$ rad/sec and $\omega_n = 2\pi \times 5$ rad/sec, respectively.

According to Table 1, the mode $\lambda_1 = -109$ is a well damped mode corresponding to the generator speed $\Delta\omega_r$.

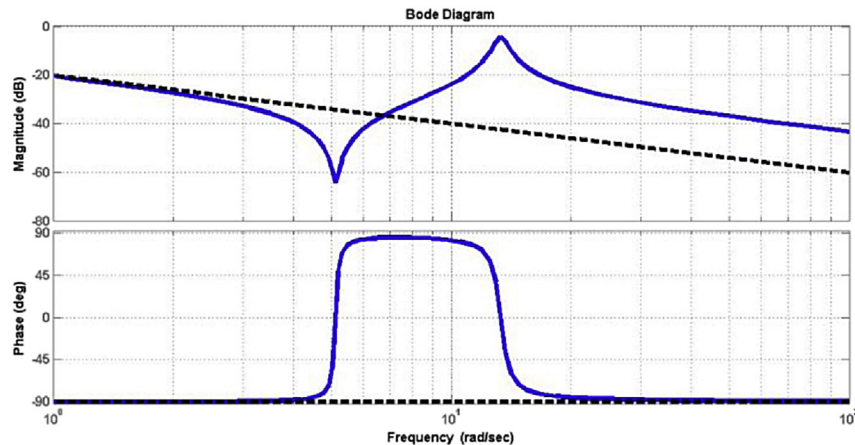


Fig. 4. Bode diagram of $\Delta\omega_r/\Delta T_e$ transfer function, solid line: two-mass model, dashed line: single-mass model.

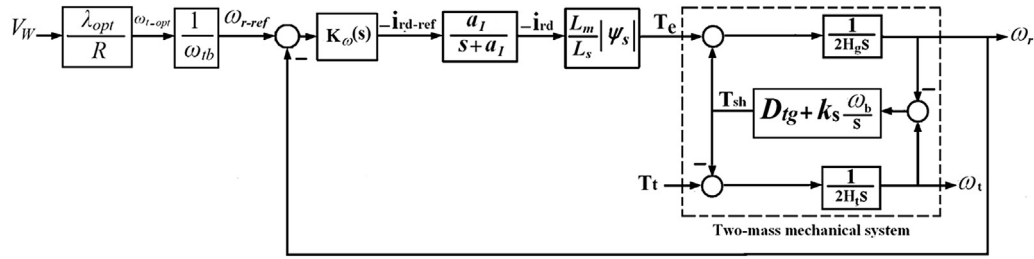


Fig. 5. Wind turbine speed control loop based on the two mass drive train model.

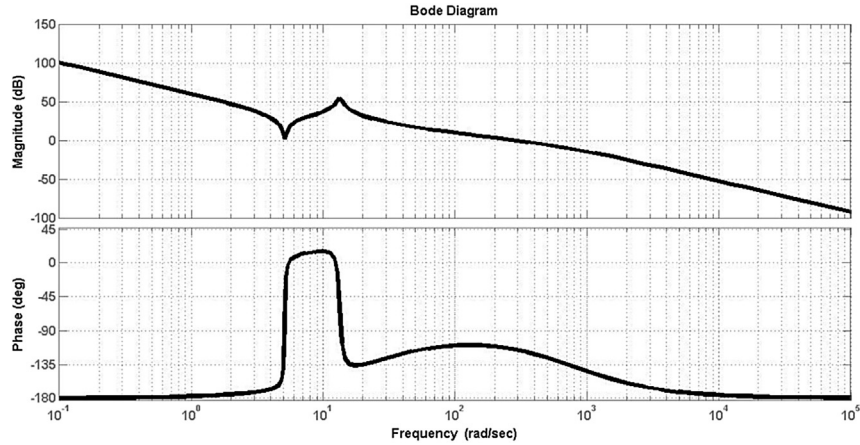


Fig. 6. Bode plot of the open loop transfer function given in (22).

Table 1

System modes, and dominant state variables for the speed control bandwidth of $\omega_n = 2\pi \times 2$ rad/sec.

System modes	Dominant state variables
$\lambda_1 = -109$	$\Delta\omega_r$
$\lambda_2 = -10.85$	Δx_4
$\lambda_{3,4} = -0.19 \pm j4.85$	$\Delta\omega_t$ and $\Delta\theta$

Table 2

System modes, and dominant state variables for the speed control bandwidth of $\omega_n = 2\pi \times 5$ rad/sec.

System modes	Dominant state variables
$\lambda_1 = -275$	$\Delta\omega_r$
$\lambda_2 = -24.7$	Δx_4
$\lambda_{3,4} = -0.08 \pm j5.07$	$\Delta\omega_t$ and $\Delta\theta$

The mode $\lambda_2 = -10.85$ is a simple damped mode corresponding to the state variable Δx_4 . This mode is approximately equal to the bandwidth of speed control loop.

The modes $\lambda_{3,4} = -0.19 \pm j4.85$ are poorly damped oscillatory mechanical modes corresponding to the turbine speed $\Delta\omega_t$ and shaft twist angle $\Delta\theta$. The corresponding damping ratio and natural frequency are $\xi = 0.039$ and $\omega_n = 4.85$ rad/sec (0.77 Hz), respectively.

Considering Tables 1 and 2, increasing the speed control bandwidth may result in the oscillatory mechanical modes with lower damping.

The speed control given in Section 2.2 is designed based on the single mass drive train model in which the turbine and shaft twist angle dynamics are ignored. Hence, speed control design based on the single mass model can result in weakly mechanical modes for lower speed control bandwidth and even unstable mechanical

modes for higher speed control bandwidth. Therefore, in the design of the speed controller, the dynamics of the shaft system should be taken into account. In this way, Section 4 deals with the speed controller design considering two mass drive train dynamics.

4. Speed control with feedforward compensation of torsional torque

As discussed in Section 3, the design of the speed controller based on the single mass model does not consider the shaft dynamics and may consequently result in stability problem and mechanical resonance vibration.

In the multi mass drive train model, there is coupling between the generator speed and dynamics of the shaft twist angle and turbine speed. The shaft torsional torque indeed reflects the effects of the shaft twist angle and turbine speed on the generator speed dynamics. Hence, feedforward compensation of the shaft torsional torque can be a simple and straightforward approach for removing the effects of shaft dynamics on the generator speed and improving the generator speed dynamics. This Section answers to the following question: “Why the simple control approach of the shaft torsional torque compensation may deteriorate the drive train dynamic response?”

Fig. 7 shows the closed loop control of the generator speed with the feedforward compensation of the shaft torsional torque T_{sh} . By feedforward compensation of the shaft torsional torque T_{sh} , the generator speed control is completely decoupled from the shaft and turbine dynamics, and the drive train dynamics are divided into two subsystems. The first subsystem is the generator speed control loop, and the second one is the shaft-turbine dynamics called internal dynamics.

With the appropriate design of the speed controller, the stability and dynamic performance of the generator speed are improved.

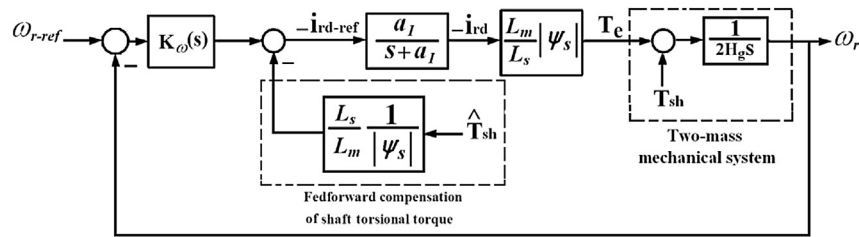


Fig. 7. Control of the generator speed with feedforward compensation of shaft torsional torque.

However, the internal dynamics of the shaft and turbine are not observable from the speed control point of view. In this way, the internal dynamics of the shaft and turbine are obtained as follows by setting the generator speed to its reference value in (18) and (20), i.e. $\Delta\omega_r = 0$ in (18) and (20).

$$\frac{d\Delta\theta_{sh}}{dt} = \omega_b \cdot \Delta\omega_t \quad (24)$$

$$2H_t \frac{d\Delta\omega_t}{dt} = \Delta T_t - D_{tg} \cdot \Delta\omega_t - k_s \cdot \Delta\theta_{sh} \quad (25)$$

Considering (24)–(25), the transfer function from the turbine torque, ΔT_t , to the shaft twist angle, θ_{sh} , is given by

$$\frac{\theta_{sh}}{T_m} = \frac{\omega_b}{2H_t s^2 + D_{tg}s + k_s \omega_b} \quad (26)$$

According to (26), the internal dynamics of the shaft-turbine is a poorly damped system with damping ratio of $\xi = \frac{D_g}{2\sqrt{k_s \omega_0}}$ and undamped natural frequency of $\omega_0 = \sqrt{\frac{k_s \omega_0}{2H_f}}$. For the DFIG test system with the parameters of [Appendix A](#), ξ and ω_0 are 0.07 and 5.12 rad/sec, respectively. $\square$

According to Section (3.2), $\omega_0 = \sqrt{\frac{k_t \omega_n}{m}}$ is exactly equal to the anti-resonant frequency ω_{02} . This means that under compensation of the shaft torsional torque, the natural resonant frequency of the internal dynamics is exactly the same as the anti-resonant frequency ω_{02} , and the damping ratio is relatively low.

In the following, we examine the wind turbine response under compensation of the shaft torsional torque. The system under study is shown in Fig. 1, and the DFIG based WT parameters are given in Appendix A.

Fig. 8 depicts the time responses of the shaft torsional torque T_{sh} and electromagnetic torque T_e . In Fig. 8, the wind speed is initially 11 m/sec and is changed to 12 m/sec at $t = 2.5$ s.

According to Fig. 8, undamped oscillations with frequency of $\omega_0 = \sqrt{\frac{k_s \omega_n}{2H}}$ are appeared on the shaft torsional torque and electromagnetic torque after the step change of the wind speed. This may result in high stress on the drive train dynamics. Hence, in the case of the shaft torsional torque compensation, even though the generator speed may be well damped, however, the weakly damped oscillatory modes of the internal dynamics results in poorly damped oscillations of the torsional and electromagnetic torque during the transient conditions. In this case, the control law of torsional torque compensation that stabilizes the generator speed, but does not take into account the internal dynamics, may result in a poorly damped whole system.

5. Speed controller design based on the two-mass drive train model

As given in (22), the two mass drive train model contains oscillatory pole and zero with the undamped natural frequencies ω_{01} and ω_{02} known as resonant and anti-resonant frequencies.

As stated in Section 3, in order to damp low-frequency oscillations of the drive train, the speed controller should be properly designed based on the two-mass drive train model.

The speed regulator is designed to attain the following specifications: stable closed loop system with adequate stability margin, high low-frequency loop gain to have zero tracking error and better low-frequency disturbance rejection, and attenuation of the open-loop gain in the frequency region near the resonant frequency.

To realize the above mentioned specifications, the following guidelines for the speed controller are considered:

- (a) The speed controller is considered to be a PI regulator as $K_{\omega}(s) = k_{p-\omega} + k_{I-\omega}/s$ with the corner frequency $\omega_{3db-Pi} = \frac{k_{I-\omega}}{k_{p-\omega}}$, to have zero tracking error. The corner frequency of the PI controller should be sufficiently far below the anti-resonant frequency such that the positive phase of the controller zero is added at the lower frequencies to avoid of low phase margin. This results in a closed loop system with high low-frequency loop gain and sufficiently low bandwidth less than the resonant and anti-resonant frequencies. In this study, the PI corner frequency is selected as $\omega_{3db-Pi} = \frac{1}{25} \omega_{02}$, where for the DFIG system under study, ω_{02} is 5.12 rad/sec and the PI corner frequency is selected as $\omega_{3db-Pi} = \frac{1}{25} \omega_{02} = 0.2 \text{ rad/sec}$.
- (b) To have a stable control loop, the open loop crossover frequency should be far below the anti-resonant frequency; otherwise the phase margin in the vicinity of this frequency may be too low. In this study, the proportional gain of the PI regulator, $k_{p-\omega}$, is selected such that the crossover frequency (ω_c) of the overall open loop transfer function be equal to 2 rad/sec which is less than 1/2 of the anti-resonant frequency.

For setting the cross over frequency equal to 2 rad/sec, i.e. $\omega_c = 2$ rad/sec, it is required that the magnitude of the overall open loop transfer function, given in (22), at $\omega_c = 2$ rad/sec be equal to 1, i.e. $|G_{op-\omega}(j\omega_c)| = 1$. Hence, by setting $|G_{op-\omega}(j2)| = 1$ and considering $\frac{k_{l-\omega}}{k_{p-\omega}} = 0.2$, the value of $k_{p-\omega}$ is computed equal to 19.8.

- (c) In order to improve the dynamic performance of the drive train system, the loop gain of the speed control system should be attenuated in the frequency region near the resonant frequency. In this study, as will be shown later, we use a notch filter at the output of PI regulator to reduce the open loop gain at the resonant frequency.

In order to limit the open loop gain at the resonant frequency, the proportional gain $k_{p-\omega}$ should be sufficiently low. However, the low value of $k_{p-\omega}$ results in the slow dynamic response. Even the too low value of $k_{p-\omega}$ may result in the stability problem. Hence, a trade off is needed in selection of appropriate proportional gain $k_{p-\omega}$. In the following, we deal with the speed controller design for the DFIG system of [Appendix A](#).

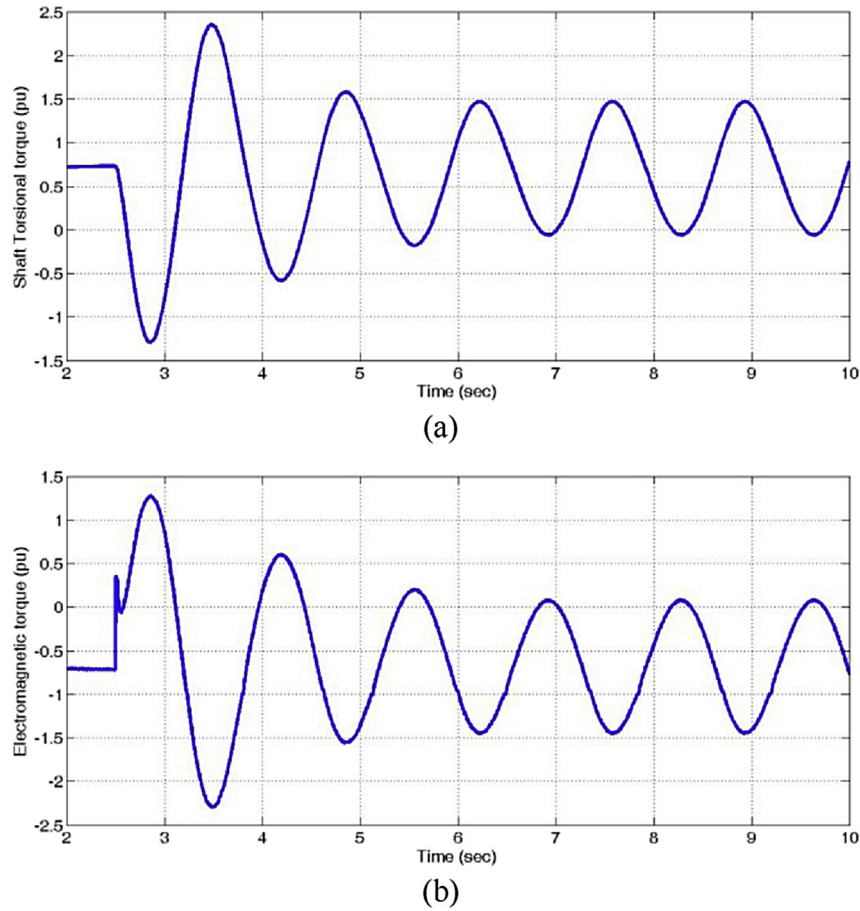


Fig. 8. Drive train response under the step change of the wind speed, (a) shaft torsional torque, (b) electromagnetic torque.

Table 3

System phase margin for different values of ω_{3db-pl} with $k_{p-\omega} = 19.8$

ω_{3db-pl} (rad/sec)	0.2	0.6	1	1.4	1.8	2.2
Phase margin (deg)	84	71	63	56	50	47

Table 3 shows the system phase margin for different values of ω_{3db-pl} with $k_{p-\omega} = 19.8$. Considering Table 3, by increasing ω_{3db-pl} , the phase margin of the system decreases.

Fig. 9 shows the bode plot of the open loop system for $k_{p-\omega} = 19.8$ and two different values of ω_{3db-pl} (0.2, 2 rad/sec). It is clear that for $\omega_{3db-pl} = 0.2$ rad/sec, the system is more stable with the higher phase margin.

Fig. 10 shows the bode plot of the open loop system for $\omega_{3db-pl} = 0.2$ rad/sec and five different values of $k_{p-\omega}$ (5, 10, 19.8, 30, 40). At $k_{p-\omega} = 40$, the phase margin is 86° representing appropriate stability margin, but the open loop gain at the resonant frequency ($\omega_{01} = 13.2$ rad/sec) is high which results in significant amplification of the torque ripple. In the case with $k_{p-\omega} = 19.8$, the phase margin is 81° , and the open loop gain at the resonant frequency is 20 dB, which is relatively high. For the case with $k_{p-\omega} = 5$, the open

loop gain at the resonant frequency is attenuated to 9.3 0 dB, but the phase margin and the system crossover frequency are reduced resulting in lower stability margin and slower dynamic response.

Hence, the higher value of $k_{p-\omega}$ results in the higher open loop gain at the resonant frequency and thus the higher amplification of the torque ripple. On the other hand, the lower value of $k_{p-\omega}$ results in the lower stability margin and the slower dynamic response.

In order to reduce the open loop gain at the resonant frequency, while having appropriate stability margin, a notch filter is placed in the feedforward loop at the output of PI regulator, as given in Fig. 11.

The notch filter, reducing the open loop gain at the resonant frequency, should provide phase lead as much as possible. Hence, the zeros of the notch filter should be on the imaginary axis, and the notch filter damping has to be small. The notch filter used for the system under study is selected as

$$G_{notch}(s) = \frac{s^2 + \omega_{01}^2}{s^2 + 2\xi_0\omega_{01}s + \omega_{01}^2} \quad (27)$$

According to Fig. 11, the overall loop gain of the speed control loop is given by

$$G_{op-\omega} = \left(k_{p-\omega} + \frac{k_{I-\omega}}{s}\right) \cdot G_{notch}(s) \cdot \left(\frac{\alpha_I}{s + \alpha_I}\right) \cdot \left(\frac{L_m}{L_s} \psi_{sd0}\right) \cdot \left(\frac{1}{2(H_g + H_t)s} \frac{2H_t s^2 + D_{tg}s + k_s \omega_b}{H_g + H_t} \frac{2H_g H_t s^2 + D_{tg}s + k_s \omega_b}{H_g + H_t}\right) \quad (28)$$

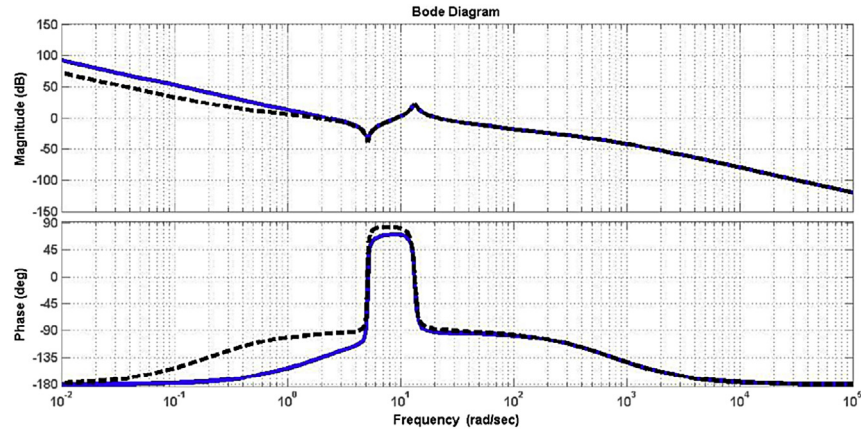


Fig. 9. Bode plot of the open loop system for $k_{P-\omega} = 19.8$ and two different values of ω_{3db-Pi} , $\omega_{3db-Pi} = 0.2$ rad/sec (solid line), $\omega_{3db-Pi} = 2$ rad/sec (dashed line).

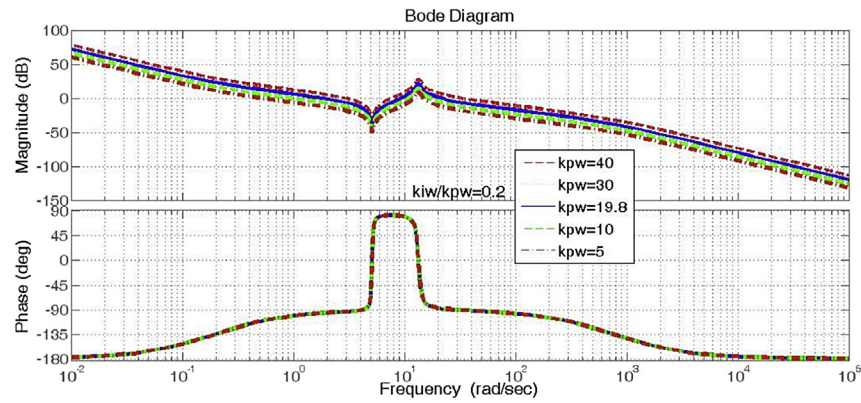


Fig. 10. Bode plot of the open loop system for $\omega_{3db-Pi} = 0.2$ rad/sec and five different values of $k_{P-\omega}$.

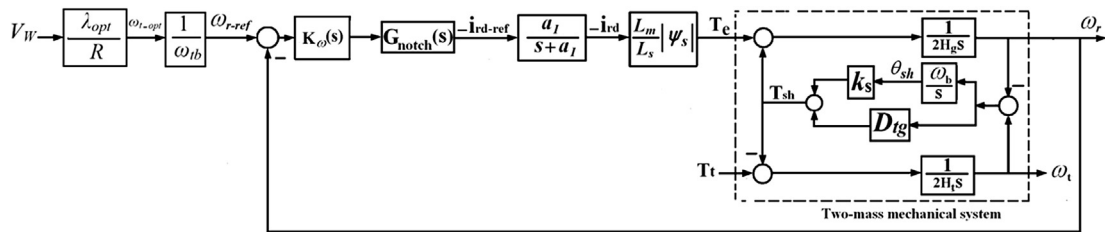


Fig. 11. Speed control loop with PI regulator and notch filter.

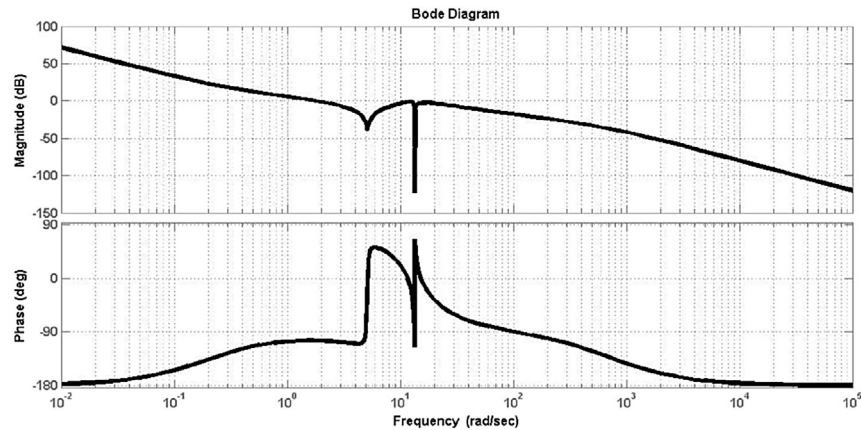


Fig. 12. Bode plot of the open loop system with the PI regulator and notch filter.

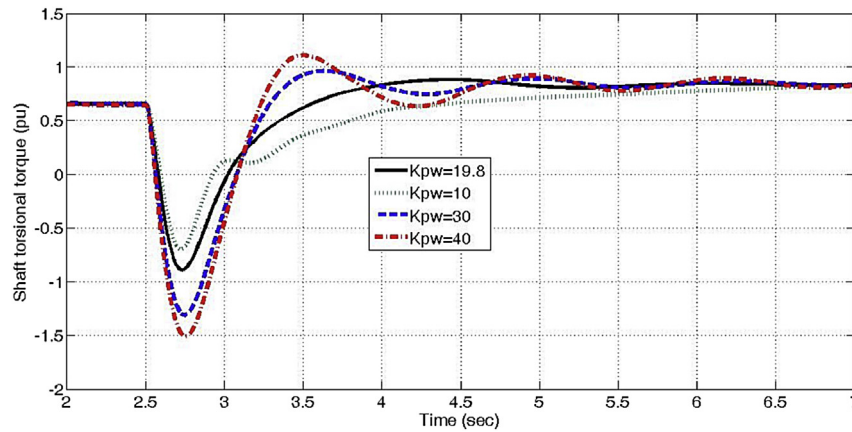


Fig. 13. Time response of the shaft torsional torque to a step change of the wind speed for different values of k_{pw} with $\omega_{3db-pl} = 0.2$ rad/sec

Fig. 12 shows the bode plot of the loop gain, given in (28), in which the crossover frequency is 2 rad/sec, and the gain at the resonant frequency (13.2 rad/sec) reaches the value of -7 dB which is below the 0 dB.

6. Time domain simulation

In this section, dynamic response of the drive train dynamics

against the step change of wind speed and aerodynamic power fluctuation is examined. The system under study is shown in Fig. 1 with the corresponding parameters given in Appendix.

6.1. Dynamic response under step change of wind speed

Fig. 13 shows the time response of the shaft torsional torque to a step change of the wind speed in which the wind speed is initially 10 m/sec and is changed to 12 m/sec at $t = 2.5$ s. In Fig. 13, ω_{3db-pl} is set to $\omega_{3db-pl} = 0.2$ rad/sec, and then performance of the control

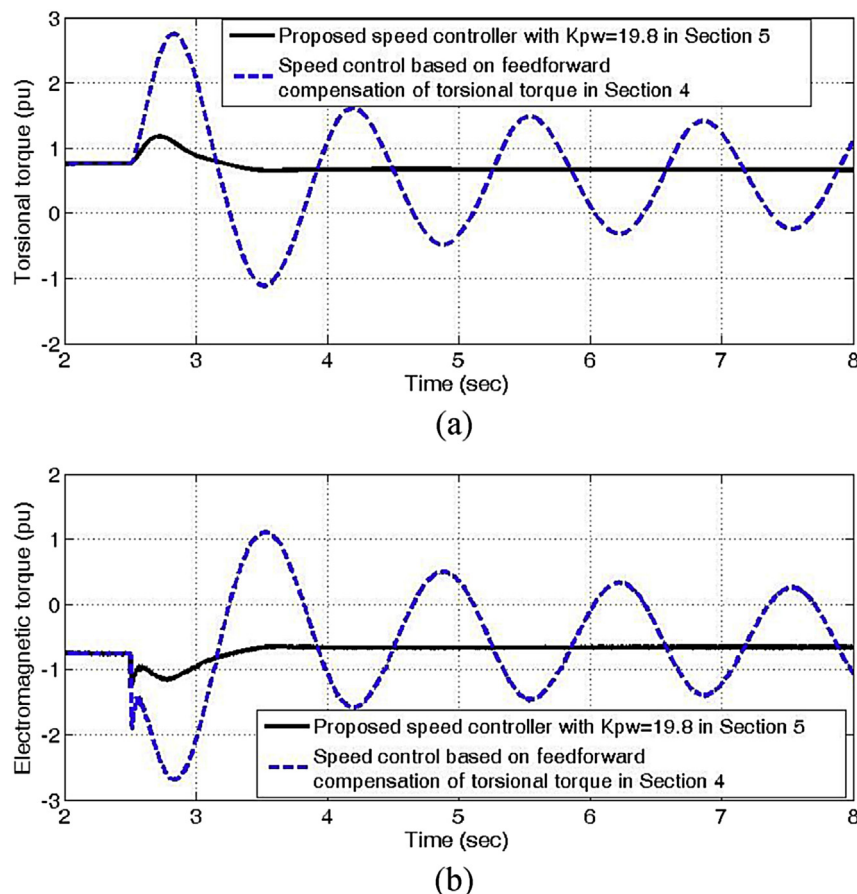


Fig. 14. Drive train response under the step change of the wind speed, (a) torsional torque, (b) electromagnetic torque.

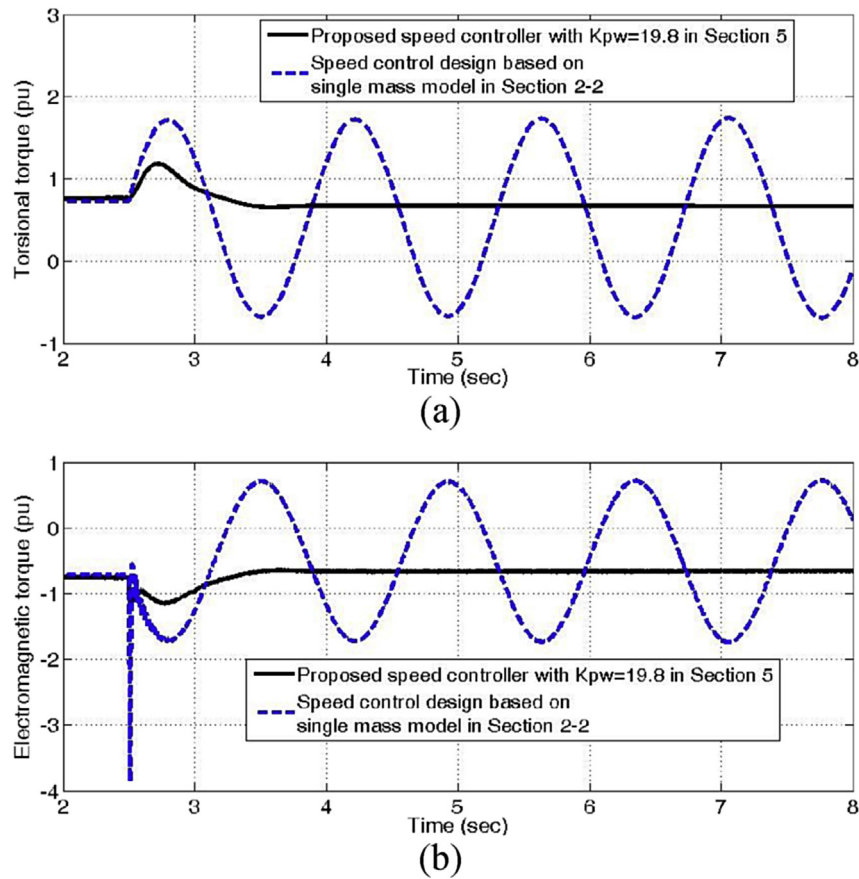


Fig. 15. Drive train response under the step change of the wind speed, (a) torsional torque, (b) electromagnetic torque.

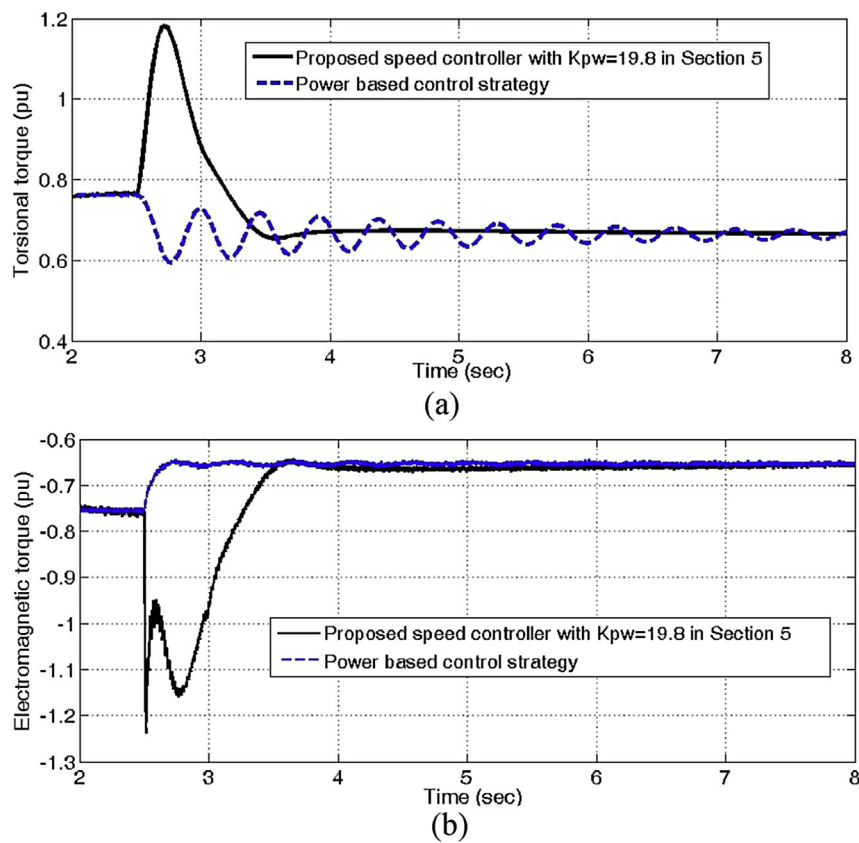


Fig. 16. Drive train response under the step change of the wind speed, (a) torsional torque, (b) electromagnetic torque.

system for different values of $k_{P-\omega}$, i.e. $k_{P-\omega}=10, 19.8, 30$ and 40 , is examined.

In other words, in Fig. 13, four distinct cases are considered: the first case associated with the proposed speed controller (joint use of the notch filter and PI regulator with $\omega_{3db-pl} = 0.2$ rad/sec and $k_{P-\omega} = 19.8$), the second case related to the PI controller with $\omega_{3db-pl} = 0.2$ rad/sec and $k_{P-\omega} = 10$, the third and fourth cases corresponding to the PI controller with $\omega_{3db-pl} = 0.2$ rad/sec and $k_{P-\omega} = 30$ and $k_{P-\omega} = 40$.

According to 13, in the case with $k_{P-\omega} = 40$, the speed control loop is faster, but the peak of the torsional torque at the instant of wind speed change reaches the higher value that may damage the mechanical system.

In the case with $k_{P-\omega} = 10$, the peak of the torsional torque at the instant of the wind speed change is lower, but the dynamic response is relatively slow. In the case with the proposed controller, the peak of the torsional torque at the instant of wind speed change is below 1 pu, which is lower than that in the cases with $k_{P-\omega} = 30$ and $k_{P-\omega} = 40$. Also, In the case with the proposed controller, the dynamic response is faster than that in the case with $k_{P-\omega} = 10$. This is because in the case with the proposed controller, the system crossover frequency has been set to an appropriate value, and the open loop gain at the resonant frequency has been attenuated below the 0 dB.

Figs. 14–16, given below, show the time responses of the shaft torsional torque and electromagnetic torque to a step change of the wind speed for different control strategies, in which the wind speed is initially 11 m/sec and is changed to 10.5 m/sec at $t = 2.5$ s. In Figs. 14–16, four different control approaches are considered: the first one associated with the control of WT in the speed control mode by the proposed speed control approach given in Section 5 (joint use of the notch filter and PI regulator with $\omega_{3db-pl} = 0.2$ rad/sec and $k_{P-\omega} = 19.8$), the second approach corresponding to control of WT in the speed control mode by the feedforward compensation of the shaft torsional torque, the third case related to the speed PI controller designed based on the single mass model (given in Section 2.2), and the fourth approach associated with the wind turbine control in the power control mode.

In Fig. 14, performance of the proposed speed controller is compared with the performance of the control approach based on the torsional torque compensation. According to Fig. 14, undamped oscillations with the frequency of 0.82 Hz (anti-resonant frequency) are appeared on the torsional torque and electromagnetic torque after the step change of the wind speed when using torsional torque compensation approach. These oscillations are not appeared when using the proposed speed controller.

In Fig. 15, performance of the proposed speed controller is compared with the performance of the speed control approach designed based on the single mass drive train model. According to Fig. 15, undamped oscillations with the frequency of 0.82 Hz (anti-resonant frequency) are appeared on the torsional torque and electromagnetic torque after the step change of the wind speed, when the speed controller is designed based on the single mass drive train model. This is in agreement with the modal analysis of Section 3.2.

In Fig. 16, performance of the wind turbine with the proposed speed controller is compared with the performance of the wind

turbine in the power control mode. According to Fig. 16, damped oscillations with the frequency of 2.1 Hz (resonant frequency) are appeared on the torsional torque and electromagnetic torque after the step change of the wind speed, when the wind turbine is controlled at the power control mode. In other words, the converter control can not provide significant damping action on the torsional oscillations when the wind turbine is controlled at the power control mode.

6.2. Dynamic response under aerodynamic power fluctuation

The aerodynamic power fluctuations in wind turbine generators are mainly related to the wind speed variations, wind shear and tower shadow effects [29]. For wind turbines with 3 blades, the wind shear and the tower shadow effects cause 3p aerodynamic torque fluctuations with frequency of ω_p , where ω_p is given by $\omega_p = \frac{3(\omega_s - \omega_2)}{n_p \cdot n_{gear}} \omega_b$, and n_p is the number of generator pole pairs, n_{gear} is the gear box ratio, ω_2 is the slip frequency in per unit, ω_s (1 p.u.) is the stator frequency, and ω_b is the base of angular frequency. The amplitude of mechanical power fluctuation is 10–20% of its average value. For the system under study with $n_p = 3$, $n_{gear} = 75$, $\omega_b = 2\pi \times 60$ rad/sec, and operating slip of $s_0 = -0.2$, the 3p oscillation frequency is obtained as $\omega_p = 6$ rad/sec, where ω_p is approximately close to the anti-resonant frequency ω_{02} .

The modal analysis carried out on the system under study shows that the drive train dynamics contain one pair of oscillatory modes.

Table 4 shows the oscillatory mechanical modes of the drive train dynamics for different values of proportional gain $k_{P-\omega}$, when $\omega_{3db-pl} = 0.2$ rad/sec. According to Table 4, by increasing $k_{P-\omega}$, the damping of the modes decreases and the frequency of the modes tends to the anti-resonant frequency which is close to the 3p oscillation frequency ω_p . Hence, for higher value of $k_{P-\omega}$, the 3p aerodynamic torque fluctuations with frequency of ω_p may be able to excite the oscillatory mechanical modes and to damage the mechanical system.

Fig. 17 depicts the time responses of the torsional torque considering aerodynamic power fluctuations due to tower shadow effects. In Fig. 17, performance of the proposed speed controller (joint use of the notch filter and PI regulator with $\omega_{3db-pl} = 0.2$ rad/sec and $k_{P-\omega} = 19.8$) is compared with the performance of other PI controllers with $\omega_{3db-pl} = 0.2$ rad/sec and $k_{P-\omega} = 30, 40, 50$.

According to Fig. 17, fluctuations of the torsional torque are lower when using the proposed speed controller. On the other hand, these fluctuations are higher for the other PI controllers with $k_{P-\omega} = 30, 40, 50$.

Figs. 18–19, given below, show the time responses of the shaft torsional torque and electromagnetic torque under aerodynamic power fluctuations originated at $t = 2.5$ s. In Figs. 18–19, three different control approaches are considered: the first one associated with the control of WT by the proposed speed control approach, the second approach corresponding to control of WT by the feedforward compensation of the shaft torsional torque, and the third case related to the speed PI controller designed based on the single mass model.

In Fig. 18, performance of the proposed speed controller is compared with the performance of the control approach based on the feedforward compensation of shaft torsional torque

Table 4

Oscillatory mechanical modes of the drive train dynamics for different values of proportional gain $k_{P-\omega}$, when $\omega_{3db-pl} = 0.2$ rad/sec

$k_{P-\omega}$	19.8	30	40	50
Oscillatory mechanical modes	$-5.9 \pm j10.9$	$-6.1 \pm j3.7$	$-2.7 \pm j4.3$	$-2.4 \pm j4.8$

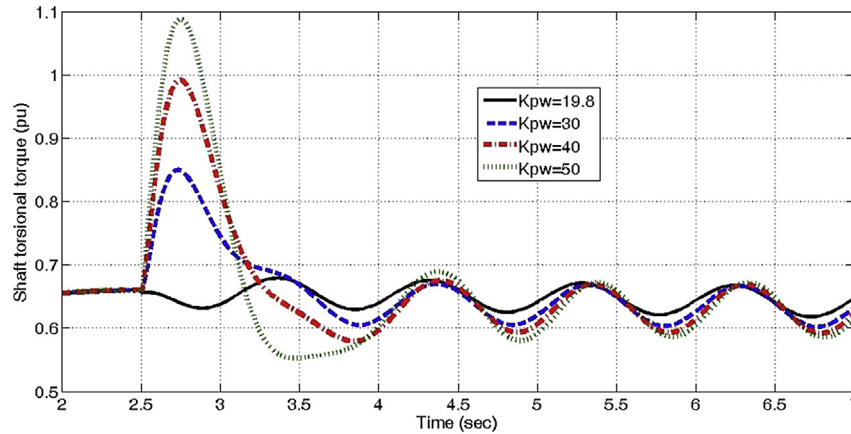


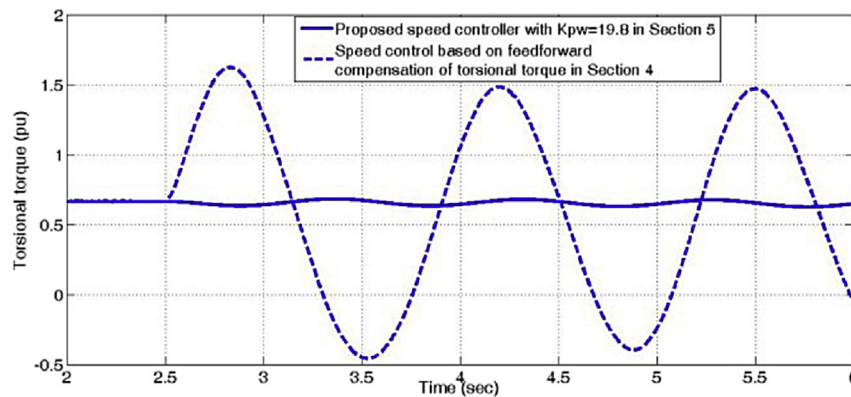
Fig. 17. Time response of the torsional torque considering aerodynamic power fluctuations due to tower shadow effects.

considering aerodynamic power fluctuations.

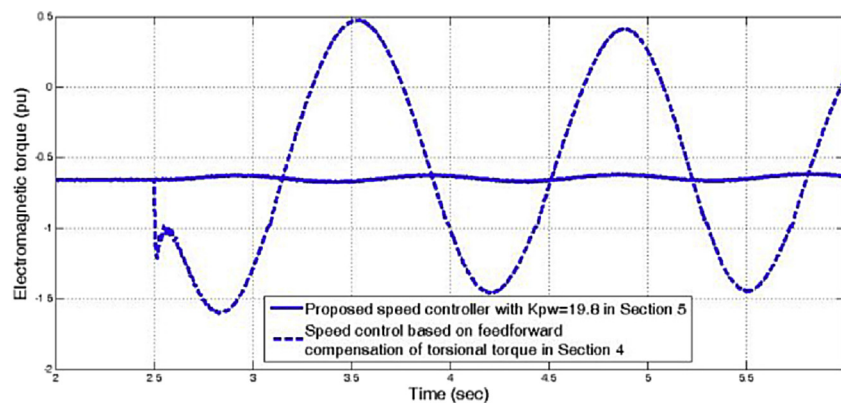
According to Fig. 18, fluctuations of the torsional and electromagnetic torque are lower when using the proposed speed controller. On the other hand, these fluctuations are higher for the control approach based on feedforward compensation of shaft torsional torque. This is because as stated in Section 4 of the manuscript, under compensation of the shaft torsional torque, the natural resonant frequency of the internal dynamics is exactly the same as the anti-resonant frequency, which is close to the 3p

oscillation frequency. Thus, in this case the 3p aerodynamic power fluctuations may excite the oscillatory mechanical modes.

In Fig. 19, performance of the proposed speed controller is compared with the performance of the speed control approach designed based on the single mass drive train model. According to Fig. 19, fluctuations of the torsional and electromagnetic torque are lower when using the proposed speed controller.



(a)



(b)

Fig. 18. Drive train response under aerodynamic power fluctuation, (a) torsional torque, (b) electromagnetic torque.

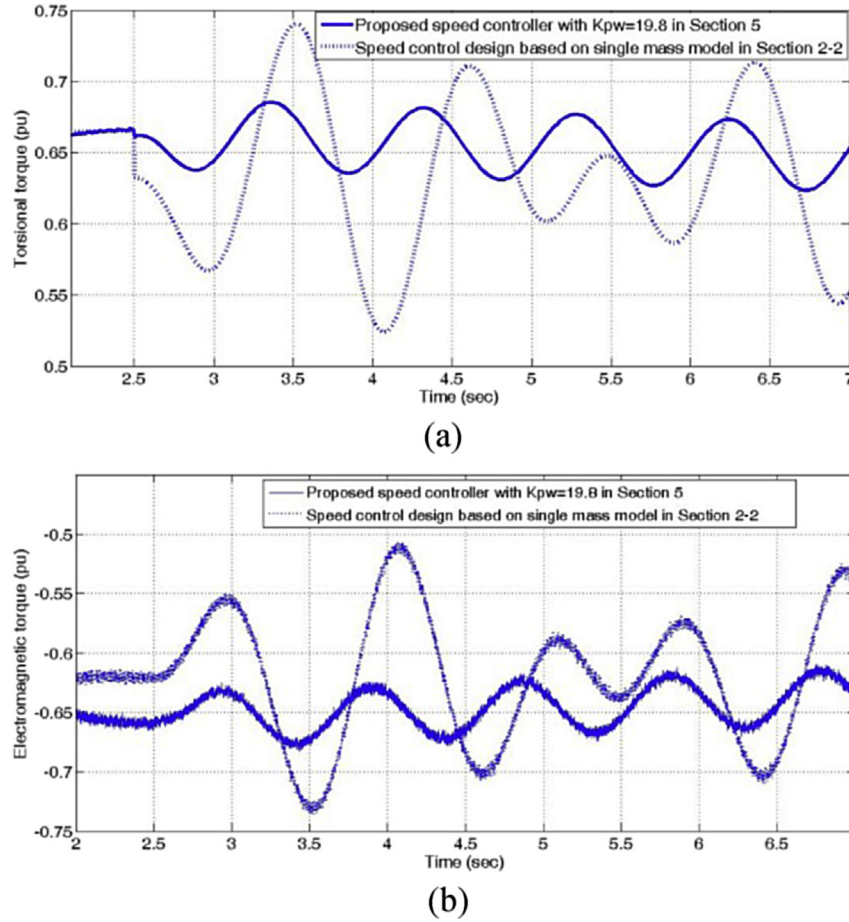


Fig. 19. Drive train response under the step change of the wind speed, (a) torsional torque, (b) electromagnetic torque.

7. Conclusion

This paper deals with the speed controller design in DFIG based wind turbines, and investigates the stability and performance of the drive train dynamics against different control strategies.

The drive train system of the wind turbine is usually represented by a two-mass model, in which the low speed mass of the turbine is connected to the high speed mass of the generator through a flexible shaft. Hence, the open loop transfer function from the electromagnetic torque to the generator speed contains a dual quadratic function representing the dynamics of flexible shaft. The dual quadratic function comprises oscillating poles and zeros with corresponding natural frequencies known as resonant and anti-resonant frequencies. The resonant and anti-resonant frequencies greatly affect the bandwidth that can be set in the speed control loop without producing instability problems.

According to the obtained results, the speed control design based on the single mass model can result in weakly damped mechanical modes for the lower speed control bandwidth and even unstable mechanical modes for the higher speed control bandwidth. This is because in this control method the dynamics of the flexible shaft is ignored.

The feedforward compensation of the shaft torsional torque is a method for improving the dynamics of the generator speed control. This control low stabilizes the generator speed, but does not take into account the internal dynamics of the shaft twist angle and turbine speed. Hence, weakly damped oscillatory modes of the internal dynamics result in poorly damped oscillations of the

torsional torque and turbine speed during the transient conditions.

In order to damp the low-frequency oscillations of the drive train, the speed controller is properly designed based on the two-mass drive train model. The proposed speed controller provides stable closed loop system with adequate stability margin, high low-frequency loop gain to have zero tracking error and better low-frequency disturbance rejection, and open-loop gain attenuation at the frequency region near the resonant frequency.

According to the simulation results, in the case with the proposed controller, fluctuations of the torsional and electromagnetic torque are lower when the wind turbine is subjected to a disturbance.

Appendix. Parameters of the system under study in Fig. 1

Appendix A. Parameters of the 1.67 MVA, 585 V, 60 Hz, DFIG-based wind turbine

$$\begin{aligned}
 V_{base} &= 585 \text{ V} \quad S_{base} = 10 \text{ MVA} \quad f_{base} = 60 \text{ Hz} \quad \omega_s = 1 \text{ pu} \\
 \omega_b &= 2\pi f_b = 377 \text{ rad/sec} \quad R_s = 0.00706 \text{ pu} \quad R_r = 0.005 \text{ pu} \\
 L_s &= 3.07 \text{ pu} \quad L_r = 3.056 \text{ pu} \quad L_m = 2.9 \text{ pu} \quad H_g = 0.75 \text{ sec} \\
 H_t &= 4.3 \text{ sec} \quad k_s = 0.6 \text{ p.u./elec.rad} \quad D_{tg} = 1.2 \text{ pu}
 \end{aligned}$$

Appendix B. Parameters of the rotor current PI-controller

$$k_{P-I_r} = 0.64 \quad k_{I-I_r} = 8.4$$

Appendix C. Parameters of the grid-side filter interfaced between GSC and grid

$$L_g = 0.15 \text{ pu} \quad R_g = 0.15/100 \text{ pu}$$

Appendix D. Parameters of the GSC current PI-controller

$$k_{P-I_{GSC}} = 0.3 \quad k_{I-I_{GSC}} = 1.13$$

Appendix E. Parameters of the DC-link capacitor

$$V_{DC} = 1200 \quad C_{DC} = 10 \text{ mF}$$

Appendix F. Parameters of the DC-link voltage PI-controller

$$k_{P-DC} = 10 \quad k_{I-DC} = 300$$

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